

2025 Mock $F = ma$ Exam

25 QUESTIONS - 75 MINUTES

INSTRUCTIONS

DO NOT OPEN THIS TEST UNTIL YOU ARE TOLD TO BEGIN

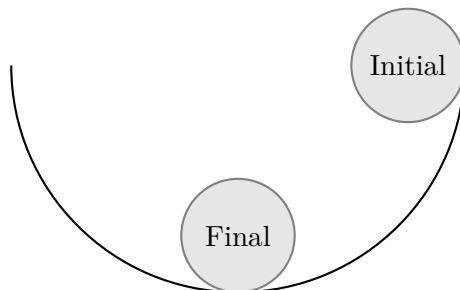
- Use $g = 10 \text{ N/kg}$ throughout this contest.
- You may write in this booklet of questions. However, you will not receive any credit for anything written in this booklet.
- Test under standard conditions, meaning that you must complete the test in 75 minutes in one sitting.
- This test contains 25 multiple-choice questions. **Your answer to each question must be marked on the [Google Forms](#) answer sheet that accompanies the test.**
- Correct answers will be awarded one point; incorrect answers and leaving an answer blank will be awarded zero points. There is no additional penalty for incorrect answers.
- A hand-held calculator may be used. Its memory must be cleared of data and programs. You may use only the basic functions found on a simple scientific calculator. Calculators may not be shared. Cell phones may not be used during the exam or while the exam papers are present. You may not use any tables, books, or collections of formulas. You must follow all of the contest rules on our [rules document](#).
- All questions are equally weighted but are not necessarily the same level of difficulty.
- This event is independent and not affiliated with, endorsed by, or sponsored by AAPT or the US Physics Team.
- **In order to maintain exam security, do not communicate any information about the questions (or their answers or solutions) on this contest until after 2:00 PM Pacific Standard Time on September 21, 2025.**

We acknowledge the following people for their contributions to this year's exam (in alphabetical order):

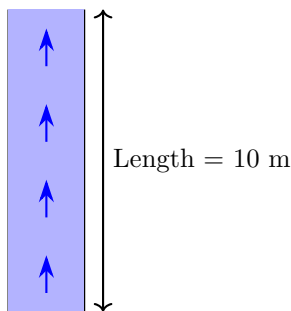
Leo Shao, Alan Wang

Special thanks to all test solvers who were able to give helpful feedback on the problems.

1. A uniform solid sphere of mass m and radius $r = \frac{R}{4}$ is released from rest at the edge of a fixed hemispherical bowl of radius R and rolls into the bowl without slipping. Find the sphere's angular displacement relative to its center when it reaches the bottom of the hemisphere.

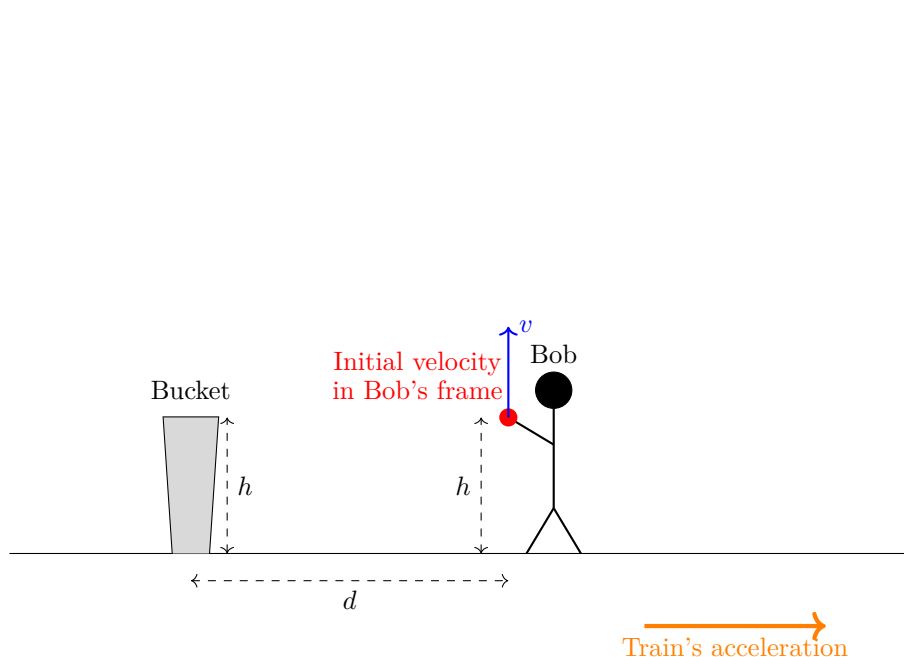


- (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) π (D) $\frac{5\pi}{4}$ (E) $\frac{3\pi}{2}$
2. A ball of mass 5 kg is launched straight up from the ground with speed v_0 . The drag force on the ball is $-av_{\text{ball}}$ for $v_{\text{ball}} \leq v_1$ and $-bv_{\text{ball}}^2$ for $v_{\text{ball}} > v_1$, where $a = 5 \text{ N}\cdot\text{s}/\text{m}$, $b = 0.50 \text{ N}\cdot\text{s}^2/\text{m}^2$, $v_1 = 10 \text{ m/s}$, and v_{ball} is the instantaneous speed of the ball. Which of the following values of v_0 result in the highest speed at which the ball first hits the ground after being launched?
- (A) 9.8 m/s (B) 9.9 m/s (C) 10.0 m/s (D) 10.1 m/s (E) 10.2 m/s
3. A system composed of many identical particles travels with a fixed total linear momentum $\vec{p}$. Which of the following conditions minimizes the total translational kinetic energy of the system?
- (A) All of the particles travel with the same speed but not the same velocity.
 (B) All of the particles travel with the same velocity.
 (C) One of the particles travels with linear momentum $\vec{p}$, with all other particles at rest.
 (D) One particle remains at rest, with all other particles traveling at an identical velocity.
 (E) None of the above.
4. Water enters a vertical cylindrical pipe at speed 10 m/s. A motor then propels it upwards at a constant velocity through the pipe. Given that the pipe has length 10 m and that the motor generates power at a rate of 500 kW, find the radius of the pipe. The density of water is $1000 \text{ kg}/\text{m}^3$. Assume that all of the power generated by the motor is transferred to the water.

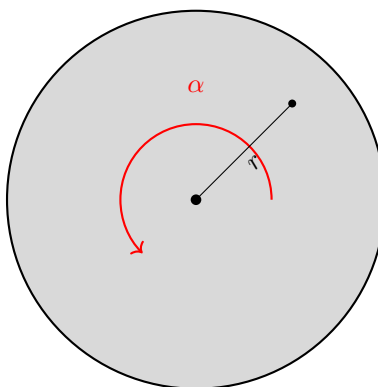


- (A) 1.26 cm (B) 1.78 cm (C) 24.8 cm (D) 39.9 cm (E) 56.4 cm

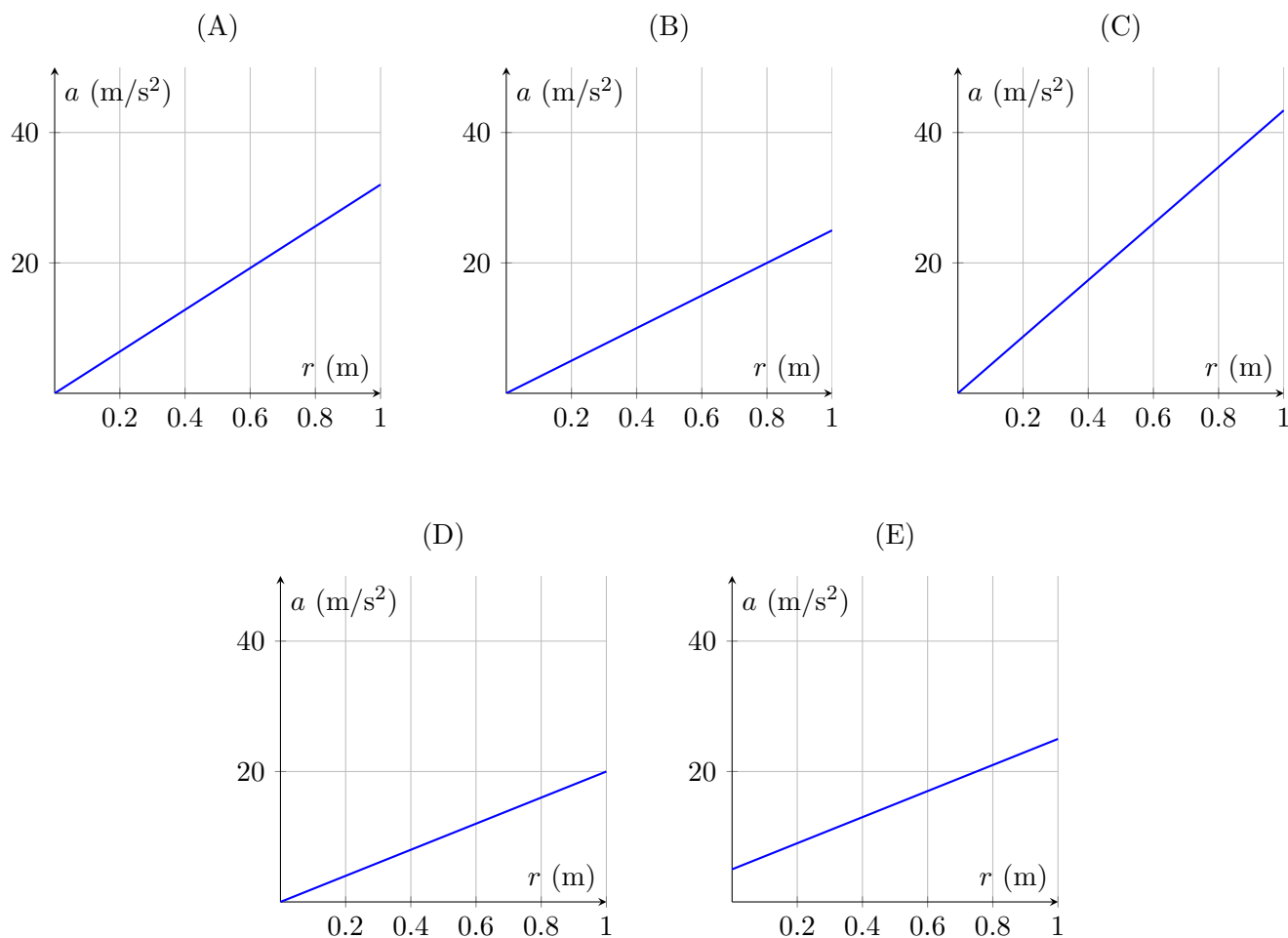
5. Bob is standing still in a very tall train that is accelerating uniformly. A small bucket, also in the train, with a height h , is a distance d directly behind Bob. In his frame, Bob tosses a small ball at height h above the ground with velocity v straight up. Find the acceleration of the train necessary for the ball to land in the bucket. Neglect air resistance, assume that the bucket does not slip relative to the floor of the train or tip over, and that the ball does not hit the roof of the train.



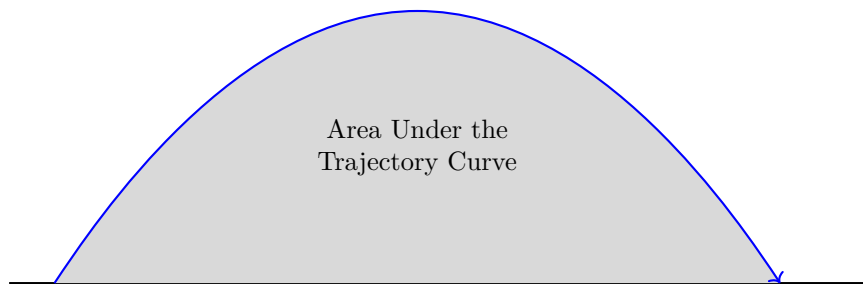
- (A) $\frac{2dg^2}{v^2}$
 (B) $\frac{dg^2}{v^2}$
 (C) $\frac{dg^2}{2v^2}$
 (D) $\frac{dg^2}{4v^2}$
 (E) The ball will never land in the bucket.
6. A solid uniform disk rotates about a pivot through its center and perpendicular to its surface with constant angular acceleration $\alpha = 20 \text{ rad/s}^2$ starting from rest at $t = 0$. Which graph could represent the acceleration a at points on the disk a distance r away from its center as a function of r at $t = 0.25 \text{ s}$?



A solid disk rotating with angular acceleration α .

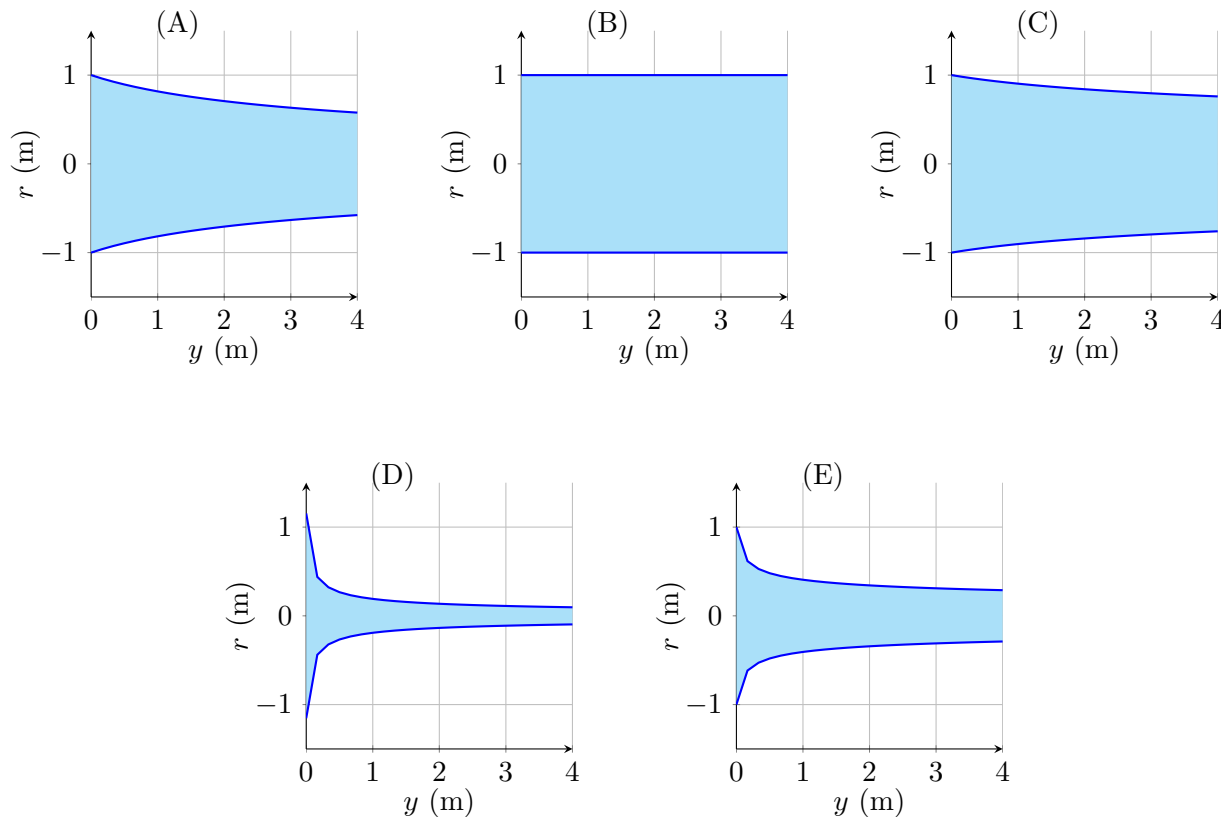


7. Projectile A is launched with an initial velocity v on a level surface at an angle θ above the horizontal on a uniform spherical planet which has mass M and radius R . Projectile B is launched with an initial velocity $4v$ on a level surface at an angle θ on a different uniform spherical planet which has mass $8M$ and radius $2R$. Let the areas under the trajectory curves of projectile A and projectile B respectively be A_A and A_B . Find A_B/A_A . Note that $v \ll \sqrt{\frac{GM}{R}}$.

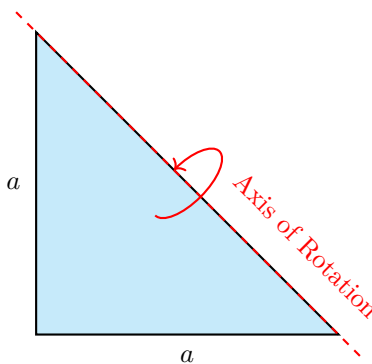


- (A) 1 (B) 4 (C) 16 (D) 64 (E) 256
8. Bob turns on a vertical faucet with a circular cross-section of radius 1 m. Which one of these diagrams best represents the shape of the stream as the water drops down? The speed of the water when it initially comes out of the faucet is 0.75 m/s. Note that the diagrams have been rotated for ease of viewing, i.e.

you may consider gravity as pointing rightward in the graphs. You may neglect the component of the water's velocity along the r direction throughout its motion.

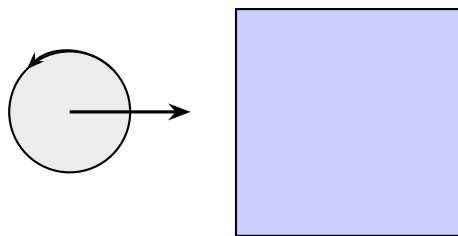


9. Calculate the moment of inertia of a planar uniform solid isosceles right triangle with legs of length a and mass m about the axis passing directly through the hypotenuse.



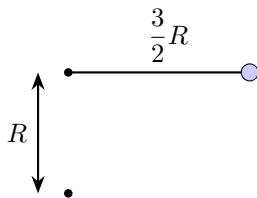
- (A) $\frac{1}{16}ma^2$ (B) $\frac{1}{12}ma^2$ (C) $\frac{1}{8}ma^2$ (D) $\frac{1}{6}ma^2$ (E) $\frac{1}{4}ma^2$

10. A solid uniform disk of mass m and radius $R = 0.8$ m lies flat on a frictionless surface. A cube of mass m and side length $l = 2$ m also lies flat on this surface at rest. As shown in the diagram, the disk is given an initial velocity $v_0 = 10$ m/s and angular velocity $\omega_0 = 30$ rad/s and collides head-on with the block so that it hits the midpoint of a side of the block. The coefficients of static and kinetic friction between the block and the disk are $\mu_s = \mu_k = 0.8$. If the disk's velocity vector at the instant after its first collision with the block is perpendicular to its initial velocity vector, find the speed of the block at that moment.



- (A) 10.00 m/s (B) 10.91 m/s (C) 11.66 m/s (D) 12.81 m/s (E) 15.62 m/s

11. A student creates a small droplet made out of an unknown liquid on a table, and slowly adds more liquid to the droplet until it breaks. If the density of the liquid is $(323 \pm 42.1) \text{ kg/m}^3$ and the surface tension between the liquid and air is $(3.75 \cdot 10^{-2} \pm 0.523 \cdot 10^{-2}) \text{ N/m}$, find the percent error in calculating the volume of the droplet right before it breaks. Assume that the values of all other relevant quantities are known to negligible uncertainty, and that the surface tensions between the table and air or liquid are 0.
- (A) 19.1% (B) 26.0% (C) 28.6% (D) 51.9% (E) 77.9%
12. Consider a flat two-dimensional Cartesian plane. A point mass is placed at rest at the origin. At $t = 0$, the particle begins accelerating. Its acceleration vector has constant magnitude $a = 10\pi \text{ m/s}^2$, rotates counterclockwise with constant angular velocity $\omega = 5\pi \text{ rad/s}$, and initially points in the x direction. Find the position of the particle at $t = 100 \text{ s}$.
- (A) (0, 100 m) (B) (100 m, 0) (C) (0, 200 m) (D) (200 m, 0) (E) (100 m, 100 m)
13. A regular uniform n -gon (a regular polygon with n sides) of side length a and mass m is held at rest on an incline angled at 25° above the horizontal. If the coefficient of static friction between the incline and the n -gon is $\mu = 0.8$, find the largest n such that the n -gon will remain in equilibrium after being released from rest.
- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8
14. Consider two infinitesimally small pivots on a frictionless wall fixed a vertical distance R apart. A massless string of length $\frac{3}{2}R$ is attached to the top pivot, and a small ball is attached to the end of the string. The ball is held perfectly horizontally and released from rest. Starting from the string's collision with the bottom pivot, find the angle that the ball swings through relative to the bottom pivot before the string loses tension.



- (A) 108°
 (B) 139°
 (C) 153°
 (D) 167°
 (E) The string does not lose tension.

15. Consider an incline of mass m angled at θ above the horizontal with a height h placed at rest on top of a frictionless floor. A small block of mass $m/10$ is held at rest at the very top of the incline and released. The block slides down the incline and comes to a complete stop just as it reaches the bottom of the incline due to friction between the incline and the block (The block does not touch the floor). Find the magnitude of the displacement of the block relative to the ground.

- (A) $\frac{h}{\sin \theta}$
 (B) $h\sqrt{1 + \left(\frac{10 \cot \theta}{9}\right)^2}$
 (C) $h\sqrt{1 + \left(\frac{9 \cot \theta}{10}\right)^2}$
 (D) $h\sqrt{1 + \left(\frac{11 \cot \theta}{10}\right)^2}$
 (E) $h\sqrt{1 + \left(\frac{10 \cot \theta}{11}\right)^2}$

16. Consider the following statements if the gravitational force between two masses m_1 and m_2 at a distance r apart was equal to $\frac{Gm_1m_2}{r}$, still being an attractive force as usual, with the units of G adjusted accordingly.

I: The gravitational field at any point inside a uniform spherical shell of nonzero mass is zero.

II: The gravitational field at any point in the plane of a uniform ring of nonzero mass, with the point completely surrounded by the ring, is zero.

Consider the following statements if the gravitational force between two masses m_1 and m_2 at a distance r apart was equal to $\frac{Gm_1m_2}{r^3}$, still being an attractive force as usual, with the units of G adjusted accordingly.

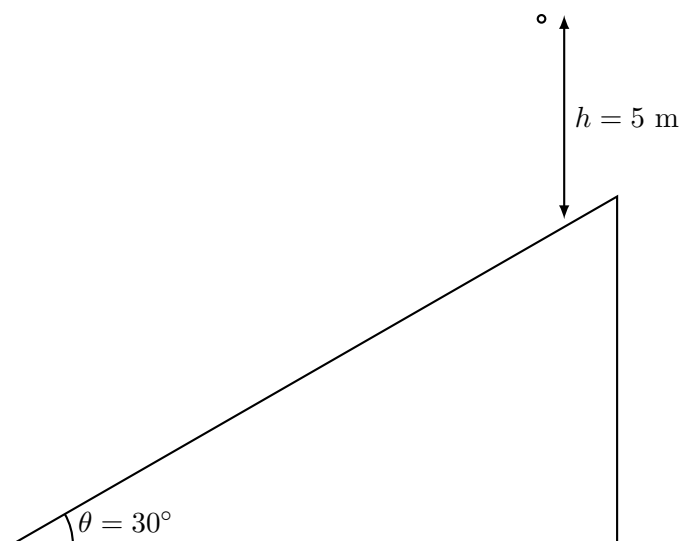
III: The gravitational field at any point inside a uniform spherical shell of nonzero mass is zero.

IV: The gravitational field at any point inside a uniform sphere of nonzero mass is zero.

Given their corresponding scenarios, which statements are true?

- (A) II only.
 (B) IV only.
 (C) I and III only.
 (D) II and IV only.
 (E) None of the statements are true.

17. A small ball is released at a height $h = 5$ m above a long fixed frictionless incline angled at 30° above the horizontal. Let point A be the point of initial contact between the small and the incline. The ball has a coefficient of restitution $\alpha = 0.90$ with the incline. Let point B be the point at which the ball stops bouncing on the incline and begins sliding down. Find the distance between point A and point B. The incline is long enough such that point A and point B are both on the incline. The diagram is not to scale.

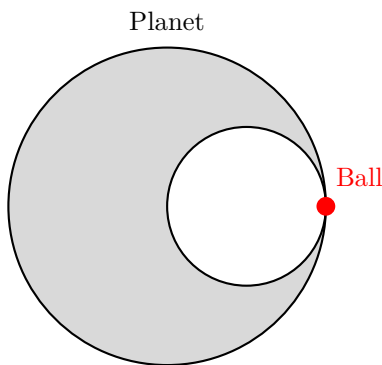


- (A) 600 m (B) 700 m (C) 800 m (D) 900 m (E) 1000 m

18. A satellite is in uniform circular motion around a planet with an orbital radius R and speed v_i . Suddenly, it applies its rocket boosters for an instantaneously short period of time, increasing its speed only in the direction of its motion. Its speed immediately after the boost is v_f . The maximum distance between the rocket and the planet in its new orbital trajectory is $5R$. Find v_f/v_i .

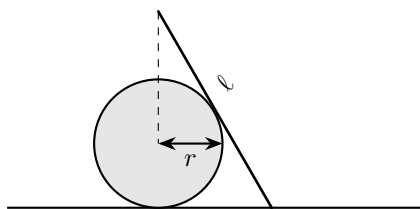
- (A) $\sqrt{2}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{4}{3}$ (D) $\sqrt{\frac{5}{3}}$ (E) $\frac{5}{3}$

19. Consider a uniform spherical planet of mass m and radius r . A spherical cavity of radius $r/2$ is cut out of the planet, which intersects the surface and the center of the planet. A small ball is placed at the intersection of the planet's surface and the cavity and released from rest. Find the time t it takes for the ball to collide with the center of the planet.



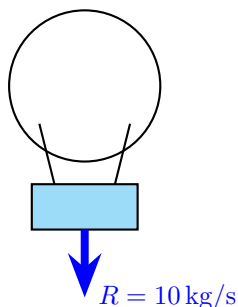
- (A) $\sqrt{\frac{r^3}{Gm}}$ (B) $\sqrt{\frac{2r^3}{Gm}}$ (C) $2\sqrt{\frac{r^3}{Gm}}$ (D) $2\sqrt{\frac{2r^3}{Gm}}$ (E) $4\sqrt{\frac{r^3}{Gm}}$

20. A cylinder is fixed to the ground. A uniform stick rests on the cylinder so that the top of the stick is positioned above the cylinder's center, as shown in the diagram. The stick makes an angle $\theta = 50^\circ$ with the horizontal. Given that the stick and cylinder have negligible friction with each other, find the coefficient of static friction μ between the stick and the ground necessary to keep the system in equilibrium. Note that the stick's corners are slightly rounded.



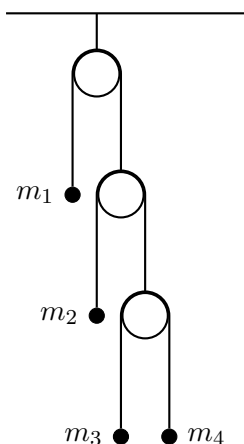
- (A) 0.386 (B) 0.499 (C) 0.538 (D) 0.564 (E) 0.839

21. A hot air balloon is initially floating in the air at rest and carries a tank of water of mass $m = 10^4$ kg in its basket. At $t = 0$, the balloon begins slowly leaking water at a constant rate of $R = 10$ kg/s. Assume that the hot air balloon's volume does not change, and neglect all energy losses due to friction or drag. The hot air balloon's mass is negligible. Which of the following answer choices is closest to the speed of the balloon at $t = 5$ s? Hint: You may use the approximation $(1 + x)^n \approx 1 + xn$ for $|xn| \ll 1$.



- (A) 0.13 m/s (B) 0.25 m/s (C) 0.38 m/s (D) 0.50 m/s (E) 0.68 m/s

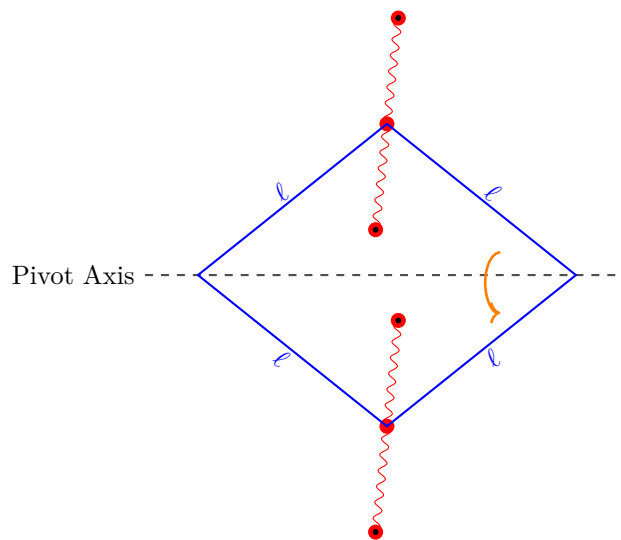
22. In the pulley system below, find the acceleration of m_1 in terms of m_2 , m_3 , m_4 , and g . All pulleys and strings are massless and frictionless. Hint: The algebra required to find the correct answer is very minimal.



- (A) $g \left(\frac{\frac{8}{m_1} - \frac{1}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
 (B) $g \left(\frac{\frac{8}{m_1} - \frac{4}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{4}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$

- (C) $g \left(\frac{\frac{16}{m_1} - \frac{2}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{16}{m_1} + \frac{2}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (D) $g \left(\frac{\frac{16}{m_1} - \frac{4}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{16}{m_1} + \frac{4}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$
- (E) $g \left(\frac{\frac{8}{m_1} - \frac{2}{m_2} - \frac{1}{m_3} - \frac{1}{m_4}}{\frac{8}{m_1} + \frac{2}{m_2} + \frac{1}{m_3} + \frac{1}{m_4}} \right)$

23. A square is constructed by attaching four uniform rigid rods, each of mass m and length l together at their endpoints. The square is then pivoted about an axis passing through two opposite corners. Four springs, each of spring constant k are attached to the other two corners such that each of the springs are perpendicular to the plane of the square and each corner has two springs attached to it, with one on each side. The springs are all initially at their relaxed length and the square is at rest. The square is then rotated by a small angle about the aforementioned axis. Find the period of small oscillations and neglect gravity.



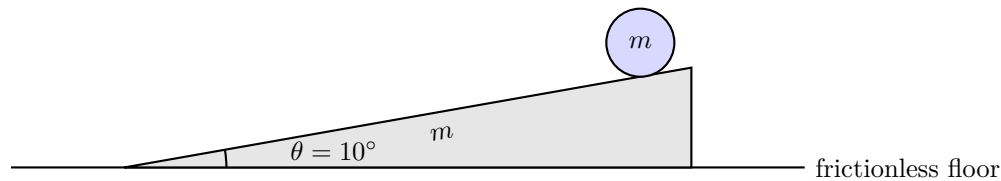
- (A) $2\pi\sqrt{\frac{m}{3k}}$ (B) $2\pi\sqrt{\frac{5m}{6k}}$ (C) $2\pi\sqrt{\frac{2m}{3k}}$ (D) $2\pi\sqrt{\frac{5m}{3k}}$ (E) $2\pi\sqrt{\frac{m}{6k}}$

24. A block is placed at rest on a frictionless floor and attached to a massless spring of spring constant k and nonzero relaxed length l . The mass is given a push so that it begins oscillating in simple harmonic motion with amplitude $A < l$. After a while, wind begins to blow in a constant direction with a constant speed directly along the block's path. The wind imparts a force $b(\vec{v} - \vec{v}_b)$ to the block, where $\vec{v}$ is the (constant) velocity vector of the wind and $\vec{v}_b$ is the (changing) velocity vector of the block. Assume that $b|\vec{v}| \ll kA$. After a long time, how does the speed of the wind impact the amplitude of the block's oscillations?



- (A) The amplitude does not change for any value of $|\vec{v}|$.
- (B) The amplitude decreases for all nonzero $|\vec{v}|$.

- (C) The amplitude increases for all nonzero $|\vec{v}|$.
- (D) The amplitude decreases for all nonzero $|\vec{v}| > x$ and increases for all nonzero $|\vec{v}| < x$, where x is a nonzero finite constant with units of m/s.
- (E) The amplitude increases for all nonzero $|\vec{v}| > x$ and decreases for all nonzero $|\vec{v}| < x$, where x is a nonzero finite constant with units of m/s.
25. A uniform cylinder of mass m is held at rest at the top of an incline angled at $\theta = 10^\circ$, also of mass m , which lies at rest on top of a frictionless floor. The cylinder is released from rest. Find the minimum coefficient of static friction μ between the cylinder and the incline such that the cylinder immediately begins rolling without slipping down the incline when it is released.



- (A) 0.0441
- (B) 0.0882
- (C) 0.176
- (D) 0.353
- (E) Achieving rolling without slipping is impossible, regardless of how large μ is.